

- **Write Once Run Anywhere (WORA)** : It means the program written in Java language can be successfully executed in other platform without any change.
- **Object Oriented Language** : It supports OOP concept i.e. Data Abstraction, Encapsulation, Inheritance and Polymorphism which enables easier and convenient program development.
- **Simple Coding** : The Java coding is simple which does not require complex instructions.
- **User Friendly Environment** : It can also be used by the novice user with fundamental programming concept.
- **Java is Robust** : The Java has automatic memory management feature and garbage collection along with strong exception handling (or checking) mechanism that makes it Java robust.
- **Secure Environment** : The Java language supports many strong in-built security features that makes the system secure from crash.

EXERCISES

Q. 1. Fill in the blanks :

1. The instructions provided to computer to solve a problem is known as a program.
2. The full form of JVM is Java virtual Machine.
3. The concept that splits a program into modules is called modular programming.
4. The portable program executed on the web browser is called applet.
5. The program that controls all the resources of a computer is called system program.

Q. 2. State True or False :

1. The Integrated Development Environment combines editor, debugger and viewer. **T**
2. The act of defining an object in many forms by the same name is called polymorphism. **T**
3. The compiler interprets a program step by step in machine language. **F**
4. The OOP has only two concepts; abstraction and inheritance. **F**
5. The JVM is a Java interpreter specially used to convert Java instructions into byte code. **F**

Q. 3. Define system programming language and application programming language.

Ans. * **SPL**

- controls the resources of a computer

- serves as a platform for running application software

* **APL**

- enables the user to perform specific tasks

- uses high level languages such as java

Q. 4. What do you mean by OOP? Name all its principles.

Ans. OOP is a programming paradigm based on the concept of objects which contain data and code

Principles are - Abstraction, Encapsulation, Inheritance and Polymorphism

Q. 5. Stats the difference between source-code and object-code.

Source Code	Object Code
* created by programmer	* created by compiler
* it is in high level code	* it is in machine level code

Q. 6. Define abstraction and polymorphism.

Ans. Abstraction: - showing essential details and hiding the rest

Polymorphism: - defining an object in many forms represented by the same name

Q. 7. Define encapsulation and inheritance.

Ans. Encapsulation: - way of combining data and methods in a single unit

Inheritance: creating a new class from an existing class. New class will have all properties of existing class as well as its own properties.

Q. 8. What do you mean by a translator? Name all its types.

Ans. It converts high level code into low level code.
Types are: Compiler and Interpreter

Q. 9. Define interpreter and compiler.

Ans. Interpreter - converts source code into Machine code line by line

Compiler - converts entire source code into Machine Code at once

Q. 10. Define applets and applications.

Ans. Applets

- * runs in a browser

- * has no access to local file system

Applications

- * runs directly under operating system

- * has full access to local file system and resources

Q. 11. What is the relation between JVM and byte code?

Ans. * byte code is interpreted by JVM

- * byte code is like source code for JVM

Q. 12. What do you mean by data and method?

Ans. * data represents values of attributes of a class

- * method acts upon the data - like modifying data